Assignment 2: Aggregates and Nested Subqueries

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Statement : Write the SQL queries using aggregates, grouping, ordering statements and nested subqueries for given statements on given schema.

A. Intermediate SQL: Aggregates and grouping and ordering

The following questions are all based on the university schema

- Find the number of instructors who have never taught any course. If the result of your query is empty, add the appropriate data (and include corresponding insert statements) to ensure the result is not empty.

Query:

```
SELECT COUNT(*) AS instructor_count
FROM instructor
WHERE id NOT IN (
      SELECT id
      FROM teaches
);
```

Output:

```
+------------------+
| instructor_count |
+------------------+
|                3 |
+------------------+
```

- Find the total capacity of every building in the university.

Query:

```
SELECT building, SUM(capacity) AS total_capacity
FROM classroom
GROUP BY building;
```

Output:

```
+----------+----------------+
| building | total_capacity |
+----------+----------------+
| Packard  |            500 |
| Painter  |             10 |
| Taylor   |             70 |
| Watson   |             80 |
+----------+----------------+
```

- Find all departments that have at least one instructor, and list the names of the departments along with the number of instructors; order the result in descending order of number of instructors.

Query:

```
SELECT dept_name, COUNT(id) AS num_instructors
FROM instructor
GROUP BY dept_name
HAVING COUNT(id) > 0
ORDER BY num_instructors DESC;
```

Output:

```
+------------+-----------------+
| dept_name  | num_instructors |
+------------+-----------------+
| Comp. Sci. |               3 |
| Finance    |               2 |
| History    |               2 |
| Physics    |               2 |
| Biology    |               1 |
| Elec. Eng. |               1 |
| Music      |               1 |
+------------+-----------------+
```

- For each student, compute the total credits they have successfully completed, i.e. total credits of courses they have taken, for which they have a non-null grade other than 'F'. Do NOT use the tot_credits attribute of student.

Query:

```
SELECT id, SUM(
      (
            SELECT credits
            FROM course
            WHERE course.course_id = takes.course_id
      )
) AS total_credits
FROM takes
WHERE grade IS NOT NULL AND grade <> 'F'
GROUP BY id;
```

Output:

```
+-------+---------------+
| id    | total_credits |
+-------+---------------+
| 00128 |             7 |
| 12345 |            14 |
| 19991 |             3 |
| 23121 |             3 |
| 44553 |             4 |
| 45678 |             7 |
| 54321 |             8 |
| 55739 |             3 |
| 76543 |             7 |
| 76653 |             3 |
| 98765 |             7 |
| 98988 |             4 |
+-------+---------------+
```

B. **Nested Subqueries** : Write the following queries for university schema.

- Find the id and title of all courses which do not require any prerequisites.

Query:

```
SELECT course_id, title
FROM course
WHERE course_id NOT IN
(
      SELECT course_id
      FROM prereq
);
```

Output:

```
+-----------+---------------------------+
| course_id | title                     |
+-----------+---------------------------+
| BIO-101   | Intro. to Biology         |
| CS-101    | Intro. to Computer Science |
| FIN-201   | Investment Banking        |
| HIS-351   | World History             |
| MU-199    | Music Video Production    |
| PHY-101   | Physical Principles       |
+-----------+---------------------------+
```

- Find the names of students who have not taken any biology dept. courses.

Query:

```
SELECT name
FROM student
WHERE id NOT IN (
      SELECT id
      FROM takes
      WHERE course_id IN (
            SELECT course_id
            FROM course
            WHERE dept_name = "Biology"
      )
);
```

Output:

```
+----------+
| name     |
+----------+
| Zhang    |
| Shankar  |
| Brandt   |
| Chavez   |
| Peltier  |
| Levy     |
| Williams |
| Sanchez  |
| Snow     |
| Brown    |
| Aoi      |
| Bourikas |
+----------+
```